LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuselage introduction, fuselage shape, structural design, detailed design layout
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuselage introduction, fuselage shape, structural design, detailed design layout
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuselage structural elements using IB diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the fuselage in an aircraft?
(b) List three main structural elements found in a typical fuselage.
(c) How does the fuselage shape differ between commercial and fighter aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Fuselage introduction and primary functions • Standard fuselage shape configurations • Monocoque and semi-monocoque structures • Skin, frames, and longerons roles • Riveting and bonding techniques	L & Ds	PPT, IB	15 min
(b)	• Detailed design layout principles • Load distribution and stress analysis • Material selection for fuselage structures • Integration of systems within fuselage • Fire protection and environmental sealing	L & Ds	PPT, PDF	20 min
(c)	• Inspection of structural element diagrams • Identification of load paths in design • Review of material properties and usage • Analysis of system integration points • Discussion on maintenance access design	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between monocoque and semi-monocoque fuselage structures.
(b) Describe how loads are distributed through the fuselage skin and frames.
(c) Why is material selection critical in detailed fuselage design layout?

12. Home Assignment and Follow Up	Prepare a brief note on how the fuselage design principles discussed in this lesson apply to the main plane structure covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to main plane, aero dynamic and structural considerations, loads on wings, basic structural requirements, role of wings as structure, positioning of components, form of wing construction	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min